

IONE: Incoherence-Oriented
Neutralisation and Extraction
for hidden effect modification in
individual participant data
meta-analysis: a simulation
study of stratification-based
extraction

Abstract

Background: In IPD meta-analysis, marginal treatment-effect estimates can be biased when hidden effect modifiers are ignored. We propose Incoherence-Oriented Neutralisation and Extraction (IONE), an exploratory diagnostic for hidden effect modification.

Methods: We simulated an IPD meta-analysis with 10 studies, a binary treatment and outcome, measured covariates carrying traces of an unmeasured modifier, and study-level variation in baseline risk and treatment prevalence. Proposed and comparator methods stratified the pooled IPD; stratum-specific risk differences were synthesised with fixed-effect and DerSimonian-Laird random-effects meta-analysis. We report ARI, C1, W_{true}/W_{est} and ATE bias reduction with Monte Carlo standard errors.

Results: In the primary scenario (n=2000, 10 studies, K=5 strata), the best method by ATE bias reduction was 1B_residual (ARI 0.032; C1 0.927; W_{true} 0.363; W_{est} 0.078). Crude ATE bias was 0.01865; stratification reduced it to 0.01345 (relative 0.279) and random-effects pooling to 0.00816 (relative 0.563). Diagnostic recovery of the true hidden structure remained modest. Sensitivity analyses showed that bias reduction depends most on sample size and effect-modification strength, and the diagnostics remained informative under non-linear covariate mappings.

Conclusions: IONE is a transparent diagnostic for hidden effect modification in IPD meta-analyses, to be reported alongside conventional models and covariate adjustment.

Keywords: individual participant data meta-analysis; evidence synthesis; heterogeneity; hidden effect modification; stratification

1. Introduction

Meta-analysis combines treatment-effect estimates from related studies [1]. IPD meta-analyses preserve participant-level covariates and can improve power for treatment-covariate interactions [2][3], but standard syntheses still estimate an average effect and may miss unmeasured effect modifiers. When the pooled population mixes subgroups with different treatment effects, a marginal summary can be biased [4][5]. We therefore frame IONE as an exploratory diagnostic that is applied before or alongside a conventional IPD meta-analysis. The term *neutralisation* means reducing the misleading influence of a marginal summary by extracting coherent subpopulations for separate analysis.

Several tools already explore heterogeneity in IPD meta-analysis, including subgroup analyses, meta-regression and mixed models with treatment-covariate interactions. These explain heterogeneity through measured covariates and study characteristics, but they do not test whether the pooled participants themselves form homogeneous subpopulations with respect to the treatment effect. As coherence diagnostics we propose C1, derived from the I^2 heterogeneity statistic [6] applied to stratum-specific log odds ratios, and W, the proportion of total conditional average treatment effect variance explained by the stratification. C1 signals between-stratum incoherence; W measures within-stratum homogeneity. In a two-stage diagnostic, the analyst first checks C1/W, and if incoherence is indicated, stratifies the pooled IPD and synthesises stratum-specific effects.

2. Methods

This simulation study follows the ADEMP framework [7].

Aims

1. Evaluate whether stratification of pooled IPD based on measured covariates can recover hidden subgroup structure.
2. Compare outcome-informed IONE methods with outcome-free and established comparators.
3. Assess whether C1 and W diagnose hidden effect modification.
4. Quantify ATE bias reduction from fixed-effect and DerSimonian-Laird random-effects synthesis of stratum-specific risk differences.
5. Identify data-generating conditions under which the diagnostics are most informative.

IPD data-generating mechanism

We simulated an IPD meta-analysis with $n=2000$ participants assigned to 10 studies. Study-level variation in baseline risk, treatment prevalence and a continuous age-like covariate was controlled by a scale parameter of 0.6. Each participant had three critical variables (Z1 continuous, Z2 binary, Z3 ordered) that affected treatment, outcome and treatment-covariate interactions, and ten measured variables (X1-X10) carrying traces of Z. A binary treatment A and binary outcome Y were generated from logistic models with Z and X main effects, a treatment effect and Z-by-A interactions. The true estimand was the population risk-difference ATE. Full algebraic details are in Additional file 1; all scenarios used fixed random seeds.

The primary scenario used $n=2000$ participants, 10 studies and $K=5$ strata. We also varied K (3, 5, 10), sample size (500, 2000, 10 000), Z-to-Y effect scale (0.5, 1.0, 2.0), Z-to-X influence scale (0.2, 0.5, 1.0), and introduced a non-linear Z-to-X mapping (quadratic and log-normal). The primary and strata-sensitivity scenarios used 50 replications; the extended robustness scenarios used 10 replications.

Stratification methods

Proposed Family 1 (outcome-informed).

- **Method 1A (Predicted probability):** logistic regression of Y on X ; stratify by quantiles of the predicted probability $\hat{p}$.
- **Method 1B (Residual):** stratify by quantiles of $\hat{Y} - \hat{p}$ from the same model.
- **Method 1C (Cross-validated):** K-fold cross-validated predictions before stratification.

Proposed Family 2 (outcome-free).

- **Method 2A (PCA):** PC1 of standardised X , stratified by quantiles.
- **Method 2B (Clustering):** K-means on standardised X with K clusters.

Comparators and baselines.

- Propensity-score quintiles [8], Gaussian mixture model on X [9], prognostic-score stratification using untreated-only $Y \sim X$ [10]. Oracle baselines stratify by true Z -space; random assignment provides a chance reference.

All quantile methods used equal-frequency strata. Logistic regression used l2 regularisation ($C=1.0$, lbfgrs). Outcome-informed methods used a 50/50 discovery/evaluation split; outcome-free methods used the full sample.

Synthesis of stratum-specific effects

Within each stratum we computed the risk difference $P(Y=1 \vee A=1) - P(Y=1 \vee A=0)$. A two-stage fixed-effect summary weighted strata by size; a DerSimonian-Laird random-effects summary estimated between-stratum variance [11]. *Neutralisation* means reducing the misleading influence of a marginal summary by extracting coherent subpopulations. Two caveats apply: discovered strata are not independent study estimates, and $C1$ is computed from log odds ratios while ATE bias is on the risk-difference scale, so $C1$ is an indirect diagnostic.

Evaluation metrics

- **ARI** [12]: agreement between estimated strata and a true-Z partition, corrected for chance.
- **C1** = $1 - I^2$ applied to stratum-specific log odds ratios [6]; lower C1 indicates stronger between-stratum heterogeneity.
- W_{true} / W_{est} : proportion of total CATE variance explained by the stratification; W_{est} is operational in real data but requires a correctly specified outcome model.
- **ATE bias reduction**: absolute differences between crude, stratified and random-effects estimates, plus relative ratios.
- Monte Carlo SE and 95% CI for every mean.

The true-Z partition is a k-means clustering of the standardised Z-space with K strata. ARI therefore measures agreement with a constructed reference.

Semi-synthetic illustrations

Five well-known Simpson-paradox examples were reconstructed as pseudo-individual records from published aggregate statistics [13][14][15][16][17]. For each example, pseudo-general variables mimicked proxies of the known confounder and the same IONE methods were applied. The candidate number of strata was selected a priori from $\{2, 3, \min(K_{true}, 4), K_{true}\}$ and K was chosen as the value maximising ARI within this set. These examples illustrate favourable and unfavourable settings for stratification; they are not validation in real IPD.

Reporting and reproducibility

The design followed ADEMP [7]; checklists are in Additional files 2 and 3. The pipeline is version-controlled; every manuscript number is read from repository CSV outputs. Code and data are at <https://github.com/bougtoir/ione-stratification-framework>.

Computational implementation

All simulations used Python 3.11. The pipeline comprises `data_generation.py`, `methods.py`, `evaluation.py`, `run_rsm_ipd_simulation.py`, `generate_summary.py` and the manuscript generator. Results are fully reproducible from the repository.

3. Results

Primary IPD scenario

Table 1 reports the primary IPD scenario ($n=2000$, 10 studies, $K=5$). Recovery of the true hidden structure was modest: the Oracle baseline achieved ARI 0.372, whereas the best non-Oracle method (1B_residual) reached ARI 0.032. C1 and W behaved as expected: methods closer to the Oracle had C1 closer to 1 and higher W, while random stratification yielded C1 0.883. W_{true} exceeded W_{est} for most methods, reflecting that the estimated CATE model captures only part of the true CATE variation.

The strongest ATE bias reduction was for 1B_residual: crude absolute risk-difference bias 0.01865, stratified 0.01345 (relative 0.279), and DerSimonian-Laird random-effects 0.00816 (relative 0.563). For comparison, random-effects pooling across the true study identifiers (i.e. a study-level meta-analysis) gave an absolute ATE bias of 0.01568 (reduction 0.00296 from crude 0.01865). Random-effects pooling reduced residual bias, showing that stratum-specific effects should be allowed to vary.

Meth od	ARI	C1	W_{true}	W_{est}	Crud e bias	Strati fied bias	RE bias	Rel reduc tion strat	Rel reduc tion RE	RE I2
1A_pr edict ed_pro	0.029	0.891	0.352	0.111	0.018 65	0.018 88	0.013 52	-0.013	0.275	0.142

b										
1B_residual	0.032	0.927	0.363	0.078	0.01865	0.01345	0.00816	0.279	0.563	0.044
1C_cv_decision	0.029	0.891	0.352	0.111	0.01865	0.01888	0.01352	-0.013	0.275	0.142
2A_PCA_cum60	0.016	0.881	0.158	0.080	0.01865	0.01883	0.01769	-0.010	0.051	0.130
2B_clustering	0.023	0.865	0.210	0.143	0.01865	0.01796	0.01651	0.037	0.115	0.146
GMM	0.005	0.895	0.050	0.096	0.01865	0.01797	0.01686	0.036	0.096	0.127
PS_propensity_score	0.005	0.881	0.050	0.059	0.01865	0.01816	0.01754	0.026	0.059	0.144
Prognostic_score	0.026	0.858	0.322	0.107	0.01865	0.01849	0.01430	0.009	0.233	0.141
baseline_oracle_kmeans	0.372	0.913	0.355	0.030	0.01865	0.01901	0.01651	-0.019	0.115	0.124
baseline_oracle_quantile	0.095	0.897	0.447	0.042	0.01865	0.01620	0.01266	0.131	0.321	0.169
baseline_random	0.000	0.883	0.004	0.004	0.01865	0.01847	0.01901	0.010	-0.019	0.117

Table 1. Primary IPD scenario ($n=2000$, 10 studies, $K=5$): means over 50 simulations.

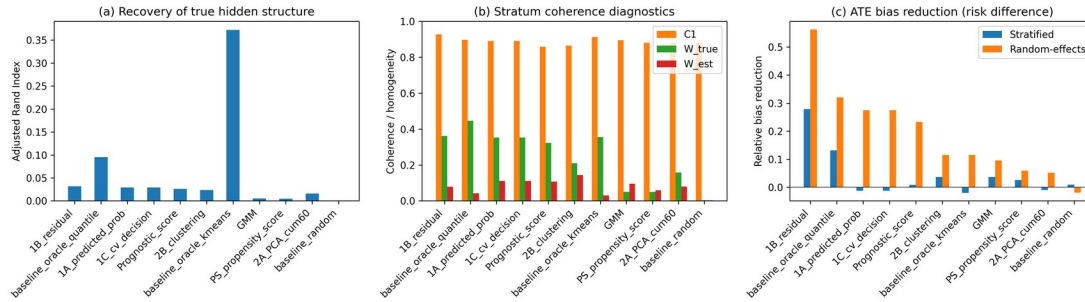


Figure 1. Primary IPD scenario: (a) ARI, (b) C1/W coherence diagnostics, and (c) ATE bias reduction by method.

Sensitivity to the number of strata

Figure 2 and Supplementary Table S1 show random-effects bias reduction across $K = 3, 5, 10$. Increasing K beyond the true dimensionality of the hidden structure did not reliably improve ATE bias reduction for data-driven methods; Oracle baselines improved because finer strata better partition the true Z -space. Stratum count should therefore be chosen conservatively and compared across several values rather than simply maximised.

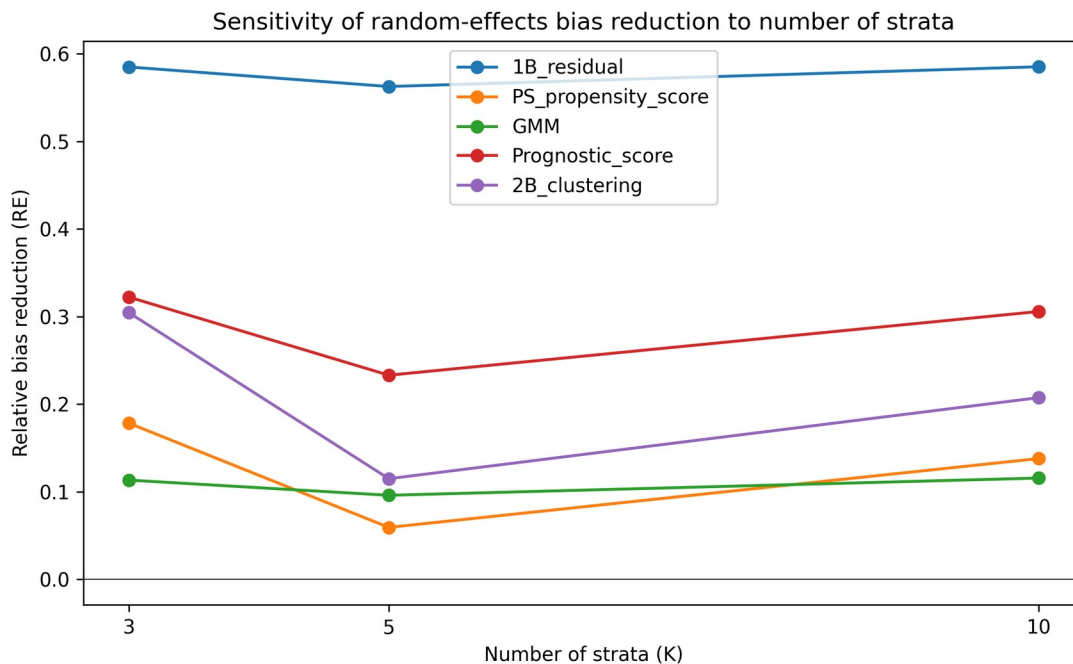


Figure 2. Random-effects ATE bias reduction as the number of strata varies.

Comparison with study-level meta-analysis

For comparison, random-effects pooling across the true study identifiers (i.e. a study-level meta-analysis) gave an absolute ATE bias of 0.01568 (reduction 0.00296 from crude 0.01865). This benchmark places the data-driven stratification results in context: even the best discovered strata reduced bias by a similar magnitude to using the true study identifier, but neither fully captures the hidden modifier.

Semi-synthetic illustrations

Five published Simpson-paradox examples were reconstructed as pseudo-individual records [13][14][15][16][17]. Table 3 reports the best non-Oracle method per dataset; high ARI occurred only when pseudo-variables strongly correlated with a low-dimensional confounder. These are illustrations, not validation in real IPD.

Dataset	Method	K	ARI	C1	W_{est}	Bias reduction
COVID-19 CFR	1B_residual	3	0.065	1.000	0.035	0.017
Israel Vaccine	2B_clustering	2	0.746	0.048	0.289	0.011
Kidney Stone	GMM	2	0.966	0.830	0.167	0.094
Smoking Mortality	1A_predicted_prob	7	0.498	1.000	0.161	0.114
UC Berkeley	PS_propensity_score	3	0.084	0.284	0.177	0.031

Table 3. Best semi-synthetic illustration result per dataset (oracle baselines excluded).

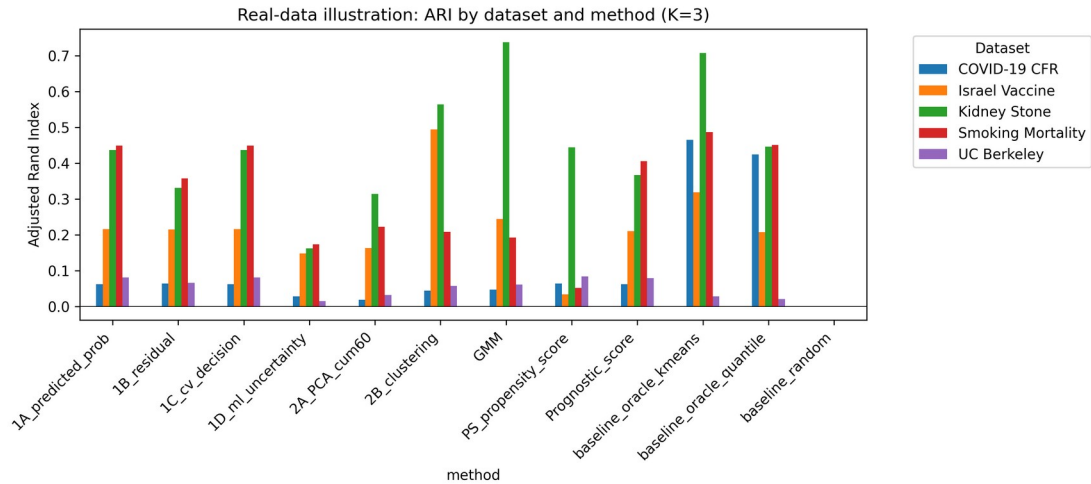


Figure 3. Semi-synthetic illustration: ARI by dataset and method.

Sensitivity to sample size

Figure 4 and Supplementary Table S2 report $n = 500, 2000$ and $10\,000$ ($K=5$). Because extended sensitivity used only 10 replications per cell, Monte Carlo SEs are included. Crude bias declined with n ; the largest relative reductions for leading methods occurred at $n=10\,000$. The outcome-residual method showed a floor near 0.008-0.009, so its relative reduction decreased with n .

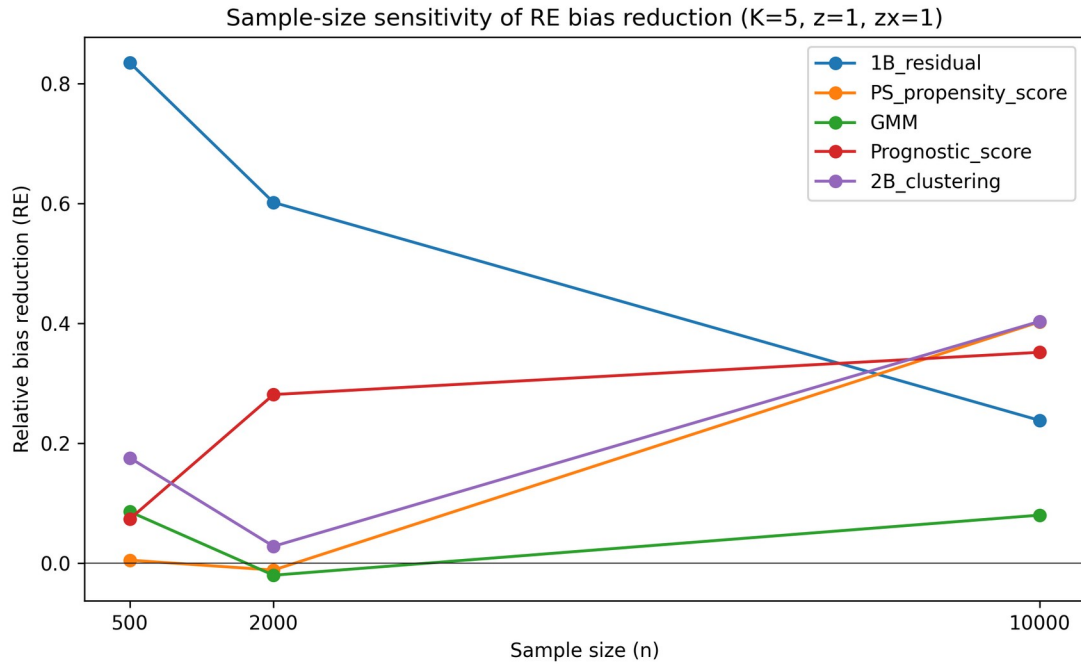


Figure 4. Sample-size sensitivity of random-effects ATE bias reduction (K=5).

Sensitivity to Z-to-X influence strength

Supplementary Table S3 reports Z-to-X influence scale = 0.2, 0.5 and 1.0. A stronger covariate trace improved ARI for most methods, but bias-reduction gains were variable at this number of replications; the diagnostics detect statistical traces rather than recover hidden variables perfectly.

Sensitivity to Z-to-Y effect strength

Supplementary Table S4 reports Z-to-Y effect scale = 0.5, 1.0 and 2.0. Weak effect modification produced C1/W values near their nulls. As the effect increased, C1 fell, W rose and relative bias reduction improved. The outcome-residual method retained useful relative reduction even at $z=0.5$, indicating it can exploit moderate covariate traces.

Robustness to non-linear Z-to-X mappings

Table 7 compares random-effects bias under linear and non-linear Z-to-X mappings. Effects were method-dependent: outcome-residual and clustering methods improved slightly, propensity-score

and GMM changed little, and prognostic score declined modestly. No leading method was dramatically degraded, indicating that the diagnostics remain useful when measured covariates are non-linear functions of the hidden structure.

Method	Linear RE bias	Linear RE bias SE	Linear rel reducti on	Linear rel reducti on SE	Non- linear RE bias	Non- linear RE bias SE	Non- linear rel reducti on	Non- linear rel reducti on SE
1B_resi dual	0.00852	0.00057	0.602	0.313	0.00664	0.00083	0.690	0.236
PS_pro pensity _score	0.02162	0.00412	-0.011	1.230	0.02084	0.00566	0.027	0.245
GMM	0.02180	0.00547	-0.020	0.544	0.01746	0.00629	0.185	0.214
Prognos tic_scor e	0.01536	0.00531	0.281	0.230	0.01762	0.00442	0.178	0.247
2B_clus tering	0.02077	0.00573	0.028	0.514	0.01649	0.00540	0.230	0.182

Table 7. Linear versus non-linear Z-to-X mapping: random-effects ATE bias and relative bias reduction ($n=2000$, $K=5$, $z=1.0$, $zx=1.0$).

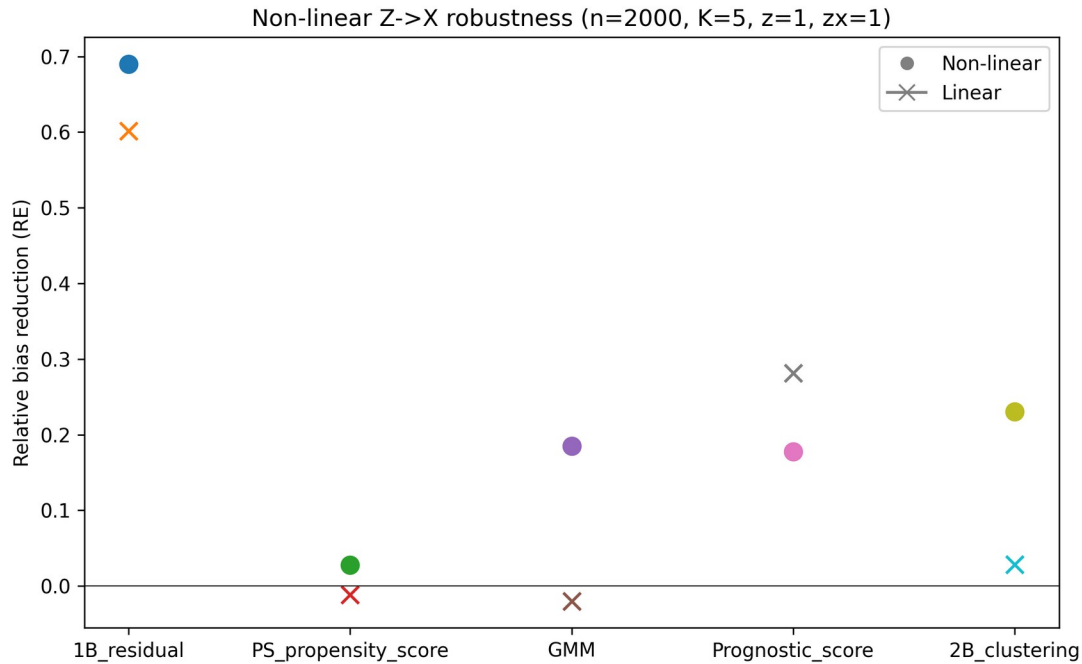


Figure 5. Non-linear Z->X robustness: random-effects ATE bias reduction (n=2000, K=5).

4. Discussion

Principal findings

IONE is an exploratory diagnostic for hidden effect modification in IPD meta-analysis. Two findings stand out. First, the coherence diagnostics C1 and W can signal hidden effect modification: methods that captured more true Z structure showed lower C1 and higher W. Second, stratification-based extraction was only conditionally successful: the Oracle baseline achieved ARI 0.372, whereas the best non-Oracle method (1B_residual) reached only 0.032. Despite modest recovery, the best data-driven method reduced crude ATE bias from 0.01865 to 0.00816 on the risk-difference scale (relative reduction 0.563) when stratum-specific estimates were pooled with DerSimonian-Laird. The extra bias reduction from random-effects pooling shows that stratum effects should be allowed to vary once incoherence is flagged.

Detection versus extraction

A two-tier interpretation is useful. Detection asks whether the pooled population is incoherent; C1 and W answer this without requiring the analyst to specify hidden subgroups. Extraction attempts to recover those subgroups and estimate stratum-specific effects. Our results show extraction is far harder than detection. IONE's practical value therefore lies mainly in detection and in partial ATE improvement, not in perfect confounding adjustment.

Relation to existing methods

Propensity-score and prognostic-score methods adjust for measured confounders [8][10]. High-dimensional propensity scores use proxy variables to improve propensity estimation [18]. Latent class analysis seeks hidden subgroups but relies on strong distributional assumptions [9]. IONE complements these: it is applied before or alongside conventional synthesis, flags incoherence, and then conventional adjustment can be applied within strata.

Strengths and limitations

Strengths include ADEMP-compliant reporting [7], a transparent pipeline that reads all numbers from repository CSVs, and a data-generating mechanism that mirrors real IPD heterogeneity.

Limitations include a single primary sample size and moderate Z-to-X trace (sensitivity analyses varied these); the true-Z partition is an operational construct, so ARI measures agreement with a constructed reference; W_{est} depends on a correctly specified outcome model; DerSimonian-Laird treats discovered strata as independent studies, which they are not; C1 is computed from log odds ratios while ATE bias is on the risk-difference scale; and the semi-synthetic examples use reconstructed aggregate data, not real individual-level records.

Implications for practice

We recommend that IPD meta-analysis reports include a coherence assessment. If $C1$ is low and W_{est} is small, the marginal effect should be interpreted cautiously and prespecified subgroups, interactions and sensitivity analyses explored. IONE should be reported alongside—not instead of—conventional models and covariate adjustment.

Future directions

Extensions include survival outcomes, network meta-analyses, Bayesian hierarchical synthesis for discovered strata, and validation on real individual-level clinical data with measured but withheld effect modifiers.

5. Conclusions

IONE (Incoherence-Oriented Neutralisation and Extraction) is an exploratory diagnostic for hidden effect modification in IPD meta-analysis. Monte Carlo simulation and semi-synthetic examples show that IONE can detect incoherence and partially reduce ATE bias when hidden variables leave strong traces in measured covariates. The diagnostics should be reported alongside conventional meta-analytic models and covariate adjustment, and should not be used as a replacement for rigorous causal inference. We recommend that coherence assessment using $C1$ and W be considered as a standard sensitivity step in IPD meta-analysis reporting.

Declarations

Ethics approval and consent to participate

Not applicable. This study used simulated data and publicly available aggregate statistics only.

Consent for publication

Not applicable.

Availability of data and materials

All code, scripts and semi-synthetic example data are available at <https://github.com/bougtoir/ione-stratification-framework>. The repository contains a requirements.txt file, fixed random seeds, and generate_summary.py that reproduces the CSV summaries from which the manuscript numbers are drawn. Running generate_summary.py, generate_ione_rsm_v3.py and generate_rsm_tables.py in Python 3.11 regenerates the main manuscript, tables and figures. An archived Zenodo DOI will be obtained before acceptance.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions (CRediT taxonomy)

Onishi Tatsuki: Conceptualisation, methodology, software, formal analysis, writing – original draft, writing – review and editing, visualisation.

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Supplementary materials

Supplementary methods, abbreviations, ADEMP/STROBE-Sim checklists, and the four extended sensitivity tables are provided in biostatistics_supplementary_v3.docx.

Artificial intelligence

Manuscript text, code and analyses were drafted or revised using OpenAI GPT-4 and GPT-4o (August 2025–August 2026) under the author's direct supervision. The author designed the study, wrote the simulation code, selected references, verified every numerical result, and approved the final scientific content.

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